User guide

Event Tree Analysis – how to model what happens after an initiating event, quantify each outcome and compare it with risk targets.

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1 · What event tree analysis is

An **event tree** is a forward-looking (inductive) model. It starts from one **initiating event** – a trip, a leak, a loss of utility – and follows how the situation develops as a series of **headings** (safety functions, barriers or conditions) either succeed or fail. Each path through the tree is an **accident sequence** ending in an **end state** (outcome) whose frequency is:

```
frequency of sequence = initiating-event frequency × Π (branch probabilities along the path)
```

Event trees and fault trees complement each other: the **event tree** shows the range of outcomes after an event; a **fault tree** explains, in detail, why one heading (for example a protection system) fails. This app can take a heading’s failure probability straight from a gate in a fault tree built with the *Fault Tree & PFD Calculator*.

Typical uses: escalation of a release (ignition, detection, isolation, deluge), consequences of a trip or loss of cooling, evaluating a new barrier or a longer test interval, quantifying the outcome categories that a LOPA or risk matrix needs.

2 · Quick start

1. **Describe the initiating event** and its frequency per year (click the empty background of the drawing, or use the *Headings & data* tab).
2. **List the headings in time order** on *Headings & data* – write each one as a success statement (“N2 system operates”, “Fire contained”) with its failure probability.

3. **Shape the tree**: on branches where a heading is irrelevant (e.g. fire fighting when there is no fire) click the branch point ● and choose *Not applicable*, or use *End the sequence here*.
4. **Define consequence categories** (tab *Consequence categories*) with tolerable frequencies and, optionally, a cost per event.
5. **Describe each end state** – click it on the drawing – and give it a category.
6. **Read the results**: category frequencies against targets, sequence list, sensitivity and uncertainty.

> The app opens with the hot DRI conveyor example (Section 12). Use *File ▸ New event tree* to start your own.

3 · Screen layout

Event tree	The drawing and a side panel for the selected item (initiating event, branch point or end state). Drag to pan, wheel or + / − to zoom, *Fit* to see everything.
Headings & data	Initiating event data; headings with labels, probabilities, fault-tree link, error factor; insert, delete and re-order headings.
Consequence categories	Outcome classes with colour, tolerable frequency target and cost per event; live status against target.
Results	Category frequencies, sequence table, sensitivity per heading, Monte Carlo uncertainty.
Fault tree links	Imported fault trees, their gate values and which headings use them.
Settings / User guide	Title, description, Monte Carlo samples; this guide.

4 · Reading the event tree drawing

- The **initiating event** is on the left with its frequency; the **headings** are the columns, left to right in time order, each showing its failure probability ("(FT)" = taken from a fault tree).
- At each **branch point** ● the path splits: the **upper branch is success**, the **lower branch is failure**; the branch labels show the probabilities. Orange labels marked * are **branch-specific** values.
- A **dashed line** means the heading is **not applicable** on that path – the path passes through without a split.
- On the right, each **sequence** has a number, an end-state description, its consequence category (coloured) and its frequency per year.

5 · Initiating event

The event that starts the sequence, with its **frequency per year**. Sources: plant trip / alarm records, maintenance history, generic data (e.g. loss of power, loss of cooling), or a fault tree with a frequency at the top. Enter an **error factor** (EF, 95th / 50th percentile) for the uncertainty run – 3 for well-documented frequencies, 5–10 for judgement.

Define it precisely: *"conveyor stops while loaded with hot DRI"* is better than *"conveyor trip"*, because the later headings depend on that condition.

6 · Headings (barriers / pivotal events)

- Put headings in the **order in which they act** in time (detection before suppression before escalation).
- Phrase each heading as a **success** statement; the probability entered is the **probability of failure** of that heading, given the path so far.
- Typical headings: prevention / recovery actions, ignition or no ignition, detection, automatic protection (SIF, deluge, inerting), operator response, mitigation (containment, fire fighting), escape / evacuation, wind direction / occupancy (conditional modifiers).
- **Success / failure labels** (e.g. “N2 works / N2 fails”) appear on the drawing.
- **Source**: *manual* value, or *fault tree* – the PFDavg of a gate in an imported fault tree (Section 9).
- **Insert / delete / move** on the *Headings & data* tab: inserting splits every path at the new heading (mark it not applicable where it does not belong); deleting keeps the success path; moving swaps the heading with its neighbour and re-arranges the branches so each combination of outcomes keeps its continuation.

7 · Branches: splits, “not applicable” and conditional probabilities

Click a branch point ● on the drawing:

- **Asked – the branch splits**: the heading is relevant on this path.
- **Not applicable – the branch passes through**: the heading cannot influence this path (e.g. there is nothing to extinguish if no ignition occurred). Changing to not applicable removes the failure branch below; changing back copies the continuation to a new failure branch.
- **End the sequence here**: all later headings become not applicable on this path.
- **Branch-specific failure probability**: overrides the heading value for this branch only. Use it for **conditional dependence** – a barrier is often less reliable after earlier failures (manual fire fighting is less likely to succeed if the automatic system failed and the fire grew; escape is harder after an explosion). Leave it blank to use the heading value.

> The sum of all sequence frequencies always equals the initiating-event frequency – a quick check that the tree is complete (shown in the side panel and checked automatically).

8 · End states and consequence categories

Every end state should get a short description of the outcome and a **consequence category**. Categories group outcomes of similar severity (e.g. from your risk matrix: minor, moderate, serious, major, catastrophic). For each category you can enter:

- **Tolerable frequency (/yr)** – the criterion from your risk matrix, LOPA target or corporate risk standard. Results show `OK` or the factor by which it is exceeded.
- **Cost per event** (any unit, e.g. k$) – gives the expected annual cost Σ frequency × cost, useful for cost-benefit comparison of improvements.

9 · Linking headings to fault trees

1. Build the fault tree for the barrier in the **Fault Tree & PFD Calculator** and save it (`.fta.json`).
2. Here: *File ▸ Import fault tree* (or the *Fault tree links* tab). The tree is stored inside the event tree file and recalculated with the same engine (exact BDD, time-averaged PFD).
3. On *Headings & data* set the heading's source to *fault tree* and choose the gate. Choose a gate that gives a **probability** – for a protection system the gate that represents "system fails on demand", not the top event if the top includes an initiating-event frequency (that would double-count the initiating event).
4. To update after changing the fault tree: *Fault tree links ▸ Replace with newer version…*; links to gates with the same internal identity are kept.

Dependencies: each heading is evaluated on its own. If two headings' fault trees share equipment (common power supply, the same control panel, the same operator), multiplying their probabilities underestimates. Model the dependency with a branch-specific probability, or put the dependent barriers into one fault tree.

10 · How the calculation works

```
F(sequence) = F(IE) × Π over split headings on the path of (1 − p) for success or p for failure
F(category) = Σ F(sequences in the category)
Expected annual cost = Σ F(category) × cost per event
```

- Headings that are *not applicable* on a path contribute a factor of 1.
- Fault-tree-linked headings use the gate's **time-averaged PFD** (PFDavg) from the fault tree engine.
- **Sensitivity**: for the selected category, each heading's probability is divided and multiplied by its error factor, and set to 0 (perfect barrier) and 1 (barrier absent), keeping all others at their base values. Branch-specific values are not changed by this.
- **Uncertainty**: the initiating-event frequency and every heading probability are sampled from lognormal distributions with mean = point value and the given EF (capped at 1); branch-specific values move with their heading (fully correlated). The run reports mean, 5th / 50th / 95th percentiles and the probability that each category exceeds its target.

11 · Reading the results

- **Category cards**: frequency per year of each category with its status against the target; expected annual cost; uncategorised frequency if any end state has no category.
- **Sequences**: S / F / – per heading, end state, category, frequency, share of the initiating event and return period (1 / F).
- **Sensitivity**: ranked by the spread between ÷EF and ×EF. The *barrier perfect* column shows the best that improving that heading can achieve; *barrier failed* shows how important it is to keep it in service (bypass, maintenance).
- **Uncertainty**: if the 95th percentile exceeds the target, the conclusion depends on data uncertainty – improve data or add margin.

12 · Worked example – hot DRI conveyor fire

Initiating event: the enclosed hot DRI transfer conveyor trips or stops while loaded, leaving hot DRI on the belt – 6 per year (illustrative).

Heading	P(failure)	Basis (illustrative)
H1 Belt restarted / hot DRI cleared before self-heating	0.25	restart or emergency discharge within the safe time
H2 No ignition – DRI does not self-heat to smouldering / belt fire	0.2	stoppage duration, DRI temperature and fines
H3 N2 inerting / extinguishing system operates on demand	≈ 0.029	**fault tree** “Hot DRI conveyor – N2 fire detection & extinguishing”, gate G-002 (PFDavg)
H4 Manual fire fighting controls the fire	0.2	response time, access to the enclosed gallery
H5 Fire contained within the conveyor gallery	0.1	fire breaks, gallery construction

Structure: if the belt is restarted (H1 success) or no ignition occurs (H2 success) the later headings are not applicable; if N2 works (H3 success) the fire is extinguished and H4–H5 are not applicable; manual fire fighting (H4) is asked only when N2 fails; containment (H5) only when manual fire fighting fails. This gives six sequences, one per consequence category C0–C5.

Results (illustrative data): C2 “fire extinguished by N2” ≈ 0.29 /yr; C3 “fire controlled manually” ≈ 7×10^{-3} /yr; C4 “major gallery fire” ≈ 1.5×10^{-3} /yr; C5 “fire spreads to transfer tower / silo” ≈ 1.7×10^{-4} /yr – above its illustrative tolerable frequency of 10^{-4} /yr by a factor of about 1.7. The sum of C3–C5 (≈ 8.6×10^{-3} /yr) matches the uncontrolled-fire frequency from the fault tree, as it should.

What it tells you: every fire sequence passes through H1–H3, so C5 is proportional to each of them. With the illustrative error factors the widest sensitivity ranges come from the judgement-based headings H2, H4 and H5 (EF 5) – better data for those would most reduce the uncertainty. The *barrier failed* column shows what the N2 system is worth: without it C5 would rise about 35-fold to ≈ 6×10^{-3} /yr, so bypass control and maintenance of the N2 system are critical. Options to bring C5 below target include improving the N2 delivery path (the fault tree shows its dominant cut sets: enclosure sealing, regulator, nozzle blockage), faster emergency discharge of hot DRI (H1), or better fire separation of the gallery (H5). Try it: set H5 to 0.05, or improve the N2 fault tree and replace it on the *Fault tree links* tab, and watch C5 change.

13 · Saving, files and export

- Automatic saving in the browser; *File ▸ Save as…* (Chrome / Edge) links an `.eta.json` file that is updated after every change; *Open file…*, *Reconnect file* after a browser restart.
- Imported fault trees are stored inside the event tree file; *Fault tree links ▸ Download .fta.json* gets them back out.
- *Export results to Excel* (summary, headings, sequences, categories, uncertainty), *Export tree picture (.svg)*, *Print report / PDF*.

14 · Good practice and limitations

Good practice

- Define the initiating event and the plant state precisely; one event tree per initiating event (or group of events with the same response).
- Order headings by time; include human actions, conditions (ignition, wind, occupancy) and mitigation, not just hardware.
- Use branch-specific probabilities for dependent barriers and record why.
- Check that each end state is described and categorised; check that sequence frequencies add up to the initiating-event frequency.
- Run sensitivity and uncertainty; document data sources in the Notes fields.

Limitations

- Binary branches (success / failure). Partial success can be modelled with an extra heading.
- Headings are independent unless you enter branch-specific probabilities; shared equipment between fault-tree-linked headings is not detected automatically.
- No timing / dynamic effects beyond the order of headings.
- Frequencies are point values with lognormal uncertainty; results are for analysis, training and design comparison – confirm decisions with your company's risk assessment procedure.

15 · Troubleshooting and FAQ

"Linked fault-tree gate not found".
The fault tree was replaced by a version without that gate, or removed. Re-select the gate on *Headings & data*; until then the manual value is used.

"The linked gate is a frequency".
You chose a gate above an initiating event in the fault tree. Choose the protection-system gate below it.

An end state shows "no category".
Click it and choose a category; uncategorised frequency is reported separately.

I changed a branch to "not applicable" by mistake.
Change it back to "asked" – the continuation is copied to a new failure branch; end-state descriptions on that new branch must be re-entered.

Can the event tree use the fault tree's top event directly?
Only if the fault tree's top is a probability (no initiating event in it). Otherwise use the gate below the initiating event.

"Reconnect file" keeps appearing.
The browser asks for write permission again after a restart – click it once.

16 · Glossary

Initiating event (IE)	Event that starts the sequence; frequency per year.
Heading / pivotal event	Safety function, barrier or condition whose success or failure determines the path.
Sequence	One path through the tree from the IE to an end state.
End state	Outcome of a sequence; grouped into consequence categories.
Conditional probability	Probability of a heading failing given what happened earlier on the path.
PFDavg	Average probability of failure on demand of a protection system (from the fault tree).
EF	Error factor: 95th / 50th percentile ratio of a lognormal uncertainty distribution.
Tolerable frequency	Maximum acceptable frequency for a consequence category (risk criterion).
Return period	1 / frequency – the average time between occurrences.

17 · References

- IEC 62502:2010 – Analysis techniques for dependability – Event tree analysis (ETA).
- CCPS – Guidelines for Hazard Evaluation Procedures (event tree analysis); Guidelines for Chemical Process Quantitative Risk Analysis.
- CCPS – Layer of Protection Analysis: Simplified Process Risk Assessment.
- NUREG/CR-2300 (1983) – PRA Procedures Guide, U.S. Nuclear Regulatory Commission.
- IEC 61511-3 – Guidance for the determination of the required safety integrity levels (risk-based methods).